

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Experiment No.2

Course Code: 23CM4121

Date: 30-08-2026

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1. TITLE

Design of a Styled Webpage Using HTML5 Graphics and CSS3 (My Styled Webpage)

2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics by designing a webpage titled "My Styled Webpage" containing a drawing (rectangle, circle, line, and name text) along with a styled header, navigation bar, content cards, and footer.

3. OBJECTIVE

- To draw basic shapes (rectangle, circle, line) and display text using HTML5 graphics.
- To design a header, navigation bar, content cards, and footer using CSS3.
- To apply background colors and a gradient background to the page.
- To use different font families and font sizes for headings and body text.
- To apply borders, border radius, margin, and padding to page elements.
- To apply box shadow and text shadow for visual depth.
- To create hover effects for navigation links and buttons.

4. THEORY

HTML5 Graphics: HTML5 allows drawing of shapes and graphics directly in the browser. Vector shapes such as rectangles, circles, and lines can be drawn using the <svg> element with tags like <rect>, <circle>, and <line>, while text can be placed on the drawing using the <text> tag. This serves the same visual purpose as the HTML5 <canvas> element — producing custom graphics without external image files — and both are commonly used together when exploring HTML5 graphics concepts.

CSS Syntax and Selectors: CSS (Cascading Style Sheets) is used to style HTML elements. A CSS rule consists of a selector (element, class, or id) and a declaration block containing property–value pairs, e.g. `nav a { color: white; }`.

Colors, Fonts and Backgrounds: Colors can be set using named colors, hex, or rgb values. The font-family property changes the typeface (e.g. Arial, Georgia, serif), and background properties set solid colors or gradients, such as `linear-gradient()`.

Borders, Border Radius, Margin and Padding: The border property draws a line around an element, border-radius rounds its corners, margin controls spacing outside an element, and padding controls spacing inside an element, between its border and content.

Box Shadow and Text Shadow: box-shadow adds a shadow effect around an element's frame, giving it depth, while text-shadow applies a similar shadow effect to text characters.

Hover Effects: The `:hover` pseudo-class applies styles to an element only while the mouse pointer is over it, commonly used to change the background or color of navigation links and buttons, combined with the `transition` property for a smooth effect.

5. ALGORITHM / PROCEDURE

- 1 Start the HTML5 document with the `<!DOCTYPE html>` declaration and set the page title to "My Styled Webpage".
- 2 Inside `<style>`, reset default margin/padding using the universal selector and set box-sizing to `border-box`.
- 3 Apply a linear-gradient background and a base font-family to the `<body>`.
- 4 Create a `<header>` with a heading and subtitle, applying text-shadow to the heading.
- 5 Create a `<nav>` bar with links, and define a `:hover` style that changes the link background with a smooth transition.
- 6 Inside a section, add a white bordered box with border-radius and box-shadow, and draw a rectangle, a circle, a line, and the name text using SVG shape elements.
- 7 Create three content cards, each styled with background color, border-radius, padding, and box-shadow.
- 8 Add a button inside each card and define a `:hover` style that changes its background color.
- 9 Create a `<footer>` with a dark background and centered copyright text.
- 10 Save the file with a `.html` extension and open it in a browser to verify the layout, gradient, shadows, rounded corners, and hover effects.
- 11 Test each styled component and the drawing individually, and verify the actual output against the expected output.

6. PROGRAM

styled-webpage.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Styled Webpage</title>
<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}
body{
    font-family:Arial, sans-serif;
    background:linear-gradient(135deg,#6a11cb,#2575fc);
    color:white;
}
/* Header */
header{
    background:#222;
    text-align:center;
    padding:25px;
}
header h1{
    font-size:40px;
    text-shadow:2px 2px 5px black;
}
header p{
    margin-top:8px;
}
/* Navigation */
nav{
    background:#111;
    text-align:center;
    padding:15px;
}
nav a{
    color:white;
    text-decoration:none;
    margin:0 20px;
    padding:8px 15px;
    border-radius:20px;
    transition:0.3s;
}
nav a:hover{
    background:#00c6ff;
}
/* Canvas Section */
.canvas-section{
    text-align:center;
    padding:40px;
}
.svg-box{
    background:white;
    width:360px;
    margin:auto;
    border:3px solid #333;
    border-radius:12px;
    padding:15px;
    box-shadow:0 8px 15px rgba(0,0,0,.4);
}
/* Cards */
.cards{
    display:flex;
    justify-content:center;
    gap:25px;
    flex-wrap:wrap;
    padding:40px;
}
.card{
    width:280px;
    background:white;
    color:#333;
    padding:20px;
    border-radius:15px;
```

```

        text-align:center;
        box-shadow:0 8px 15px rgba(0,0,0,.3);
    }
    .card h3{
        margin-bottom:10px;
        font-family:Georgia, serif;
    }
    .card p{
        margin-bottom:20px;
    }
    .card button{
        background:#2575fc;
        color:white;
        border:none;
        padding:10px 20px;
        border-radius:20px;
        cursor:pointer;
        transition:.3s;
    }
    .card button:hover{
        background:#6a11cb;
    }
    /* Footer */
    footer{
        background:#111;
        text-align:center;
        padding:18px;
        margin-top:20px;
    }
</style>
</head>
<body>
<header>
<h1>My Styled Webpage</h1>
<p>HTML & CSS Design</p>
</header>
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Services</a>
<a href="#">Contact</a>
</nav>
<section class="canvas-section">
<h2>Drawing</h2>
<div class="svg-box">
<svg width="320" height="220">
<!-- Rectangle -->
<rect x="20" y="20" width="80" height="60" fill="skyblue"/>
<!-- Circle -->
<circle cx="170" cy="50" r="30" fill="orange"/>
<!-- Line -->
<line x1="240" y1="20" x2="300" y2="90"
stroke="red" stroke-width="4"/>
<!-- Name -->
<text x="95" y="170"
font-size="24"
fill="blue">
Your Name
</text>
</svg>
</div>
</section>
<section class="cards">
<div class="card">
<h3>HTML5</h3>
<p>HTML5 provides semantic elements and multimedia support.</p>
<button>Learn More</button>
</div>
<div class="card">
<h3>CSS3</h3>
<p>CSS3 provides gradients, shadows, animations and transitions.</p>
<button>Learn More</button>
</div>
<div class="card">
<h3>Web Design</h3>
<p>Modern web pages are responsive and visually attractive.</p>
<button>Learn More</button>
</div>
</section>
<footer>

```

© 2026 My Styled Webpage | Designed by Your Name
</footer>
</body>
</html>

7. CODE EXPLANATION

Property / Tag	Description
* { margin:0; padding:0; box-sizing:border-box; }	Universal selector resets default spacing and makes width/height calculations include borders and padding.
background:linear-gradient(135deg,#6a11cb,#2575fc);	Applies a diagonal gradient background to the page body.
font-family:Arial, sans-serif / Georgia, serif;	Sets different font families for body text and card headings.
text-shadow:2px 2px 5px black	Adds a shadow behind the header heading text for a raised, glowing effect.
nav a:hover { background:#00c6ff; }	Hover effect that changes a navigation link's background color when the mouse is over it.
transition:0.3s / transition:.3s	Animates property changes (e.g. background color) smoothly over the given duration.
border:3px solid #333; border-radius:12px;	Draws a solid border around the drawing box and rounds its corners.
box-shadow:0 8px 15px rgba(0,0,0,.4)	Creates a soft drop shadow around the drawing box and cards for a raised, 3D appearance.
<svg>, <rect>, <circle>, <line>, <text>	HTML5 vector graphics elements used to draw the rectangle, circle, line, and name text.
.cards { display:flex; justify-content:center; gap:20px; }	Arranges the three content cards in a row with equal spacing between them.
.card { padding:20px; border-radius:15px; box-shadow:0 8px 15px rgba(0,0,0,.4); }	Styles each content card with inner spacing, rounded corners, and a shadow.
card button:hover { background:#6a11cb; }	Hover effect that changes a card button's background color on mouse-over.
footer { background:#111; text-align:center; padding:10px; }	Styles the footer, with a dark background, centered text, and inner spacing.

8. TEST CASES AND EXECUTION DATA

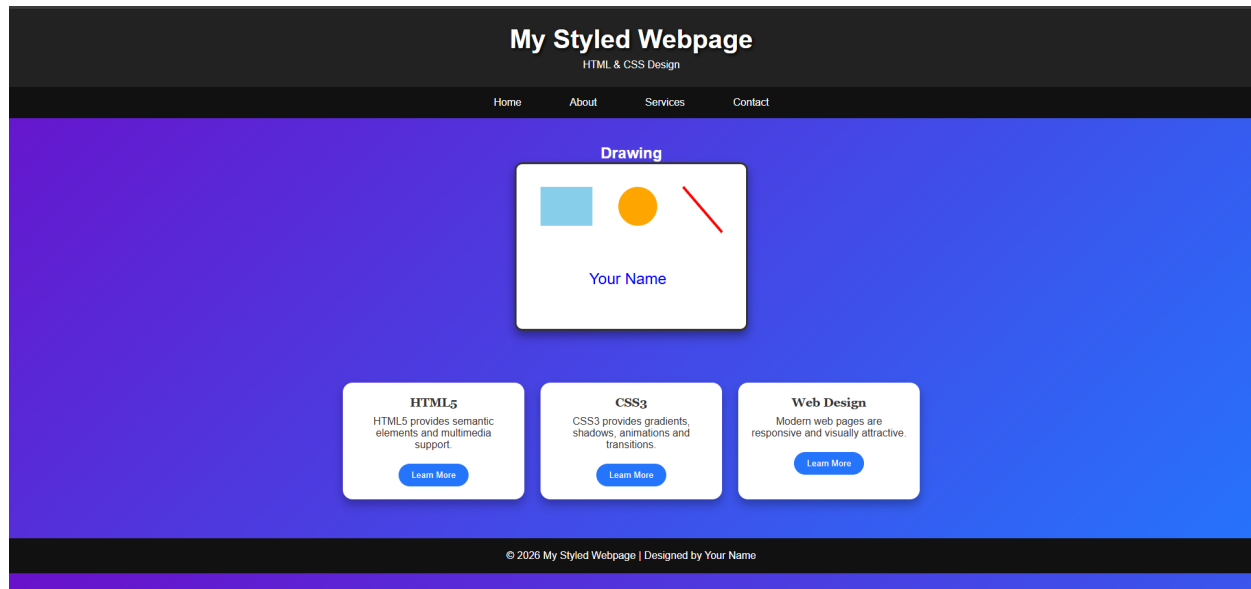
Each styled component and the drawing were verified individually by opening the file in a web browser and checking the rendered output against the expected behaviour.

Test Case	Feature Tested	Expected Result	Actual Result	Status
TC-01	Gradient background	Page body shows a diagonal purple-to-blue gradient	Gradient displayed correctly	Pass
TC-02	Header text shadow	Heading text shows a shadow for a raised effect	Shadow effect displayed correctly	Pass
TC-03	Navigation hover effect	Link background turns light blue on mouseover	Background changed smoothly on hover	Pass
TC-04	Drawing box (rectangle, circle, line, text)	Rectangle, circle, line, and name text displayed inside a display box	All shapes and text displayed correctly	Pass
TC-05	Content cards layout	Three cards displayed side by side with equal spacing	Cards aligned in a row correctly	Pass
TC-06	Card box shadow & border radius	Cards show rounded corners and a subtle drop shadow	Rounded corners and shadow displayed	Pass
TC-07	Button hover effect	Button background changes color on mouseover	Background changed on hover	Pass
TC-08	Footer styling	Footer shows dark background with centered display text	Footer displayed correctly	Pass
TC-09	Responsive card wrapping	Cards wrap to the next line on a narrow screen	Cards wrapped correctly when resized	Pass

9. OUTPUT

Output 1: Full Page Rendering

Displays the header with title and subtitle, the navigation bar, the drawing box containing the rectangle, circle, line and name text, the three styled content cards with Learn More buttons, and the footer — all rendered against the purple-to-blue gradient background.



10. RESULT

Thus, a webpage titled “My Styled Webpage” was successfully designed using **HTML5 graphics and CSS3**. A rectangle, circle, line, and name text were drawn, and the page was styled with a header, navigation bar, three content cards, and a footer, using gradient backgrounds, custom fonts, borders, border-radius, margin/padding, box shadow, text shadow, and hover effects on navigation links and buttons. The page was verified in a web browser and all sections rendered and functioned as expected.